CivicEye AI: An AI-Powered Smart Civic Issue Detection and Management System

Leveraging YOLOv8, Visual Similarity and Geospatial Intelligence for Automated Complaint Classification, Deduplication and Priority-Driven Routing

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Abstract — Urban civic governance in India faces a persistent gap between citizen reporting and municipal action: manual forms, unstructured complaints, duplicate submissions and opaque follow-ups lead to delayed redressal and citizen distrust. CivicEye AI is a full-stack, AI-augmented web platform that reduces this gap to a single photo and GPS point. Citizens upload an image of the issue; a three-task AI engine (i) classifies the issue into six civic categories using a fine-tuned YOLOv8 model, (ii) suppresses duplicates by fusing perceptual image similarity with Haversine geospatial proximity (<80 m), and (iii) computes a severity-aware priority score (0–100) that drives automatic department routing. Officers manage dept-filtered queues, update status and add remarks, while citizens receive real-time WebSocket notifications; an admin analytics layer aggregates complaint volume, resolution time and department performance. Implemented with React, Node.js/Express, PostgreSQL, Redis, Socket.IO and Docker, and demonstrated on a live Pune-centric deployment, CivicEye achieves 94.2% classification accuracy on a held-out civic test set, reduces duplicate tickets by ~80%, and completes auto-routing in under three seconds. It establishes a scalable, transparent and data-driven foundation for smart-city civic operations.

Keywords — *Smart City, Civic-Tech, Computer Vision, YOLOv8, Duplicate Detection, Geospatial Analysis, Priority Scoring, Full-Stack Development, Real-Time Systems*

I. INTRODUCTION

India's Smart Cities Mission envisions responsive, citizen-centric urban governance. Yet, routine civic issues—potholes, waste accumulation, water leakages, failed streetlights, blocked drains and damaged footpaths—continue to be reported through fragmented channels: paper registers, call centres, or at best, generic web forms that demand manual categorisation and offer no tracking. Field surveys in Pune (2023) indicate 68% citizen dissatisfaction with redressal timelines and that roughly 40% of submissions are duplicates of the same physical defect reported repeatedly within a small radius. The downstream cost is substantial: municipal staff spend an estimated 72 hours on average to route a complaint to the correct department, while administrators lack city-wide visibility into workload and service-level breaches.

Contemporary research has explored computer vision for urban defect detection and separate web portals for grievance management, but the two streams remain disconnected. Vision studies typically stop at classification and do not address deduplication, prioritisation or operational routing; portal solutions, including the Government's Swachhata application and community platforms such as FixMyStreet, provide reporting interfaces but rely on human triage. No deployed Indian system currently unifies real-time image classification, image-plus-location duplicate suppression, severity-aware auto-routing and live citizen notification within a single, scalable product.

CivicEye AI was conceived to close this loop. Its design principle is deliberately minimal for the citizen—upload one photograph and confirm GPS—while shifting intelligence to the backend. A lightweight YOLOv8-based classifier, a hybrid visual-geospatial deduplicator and a transparent priority model together decide the category, the department, and the urgency in under three seconds. Role-specific interfaces then sustain the workflow: a citizen dashboard for tracking, an officer command centre for queue management and an admin analytics suite for governance insights, all synchronised via WebSockets and backed by a PostgreSQL/Redis data layer containerised with Docker. This paper details the system's architecture, AI methodology, implementation and early evaluation, and reflects the collaborative contributions of a four-member undergraduate capstone team.

The remainder of this paper is organised as follows. Section II reviews related work and identifies the research gap. Section III presents the overall architecture. Section IV describes the AI engine and backend methodology. Section V reports implementation details. Section VI evaluates results and discusses limitations. Section VII concludes and outlines future deployment directions.

II. LITERATURE SURVEY AND GAP ANALYSIS

A. Civic Reporting Platforms

The Swachhata–MoHUA app, widely adopted for sanitation complaints, validates the demand for mobile reporting but its workflow is entirely manual: citizens select categories from dropdowns and officers re-type them into departmental spreadsheets. Analyses of its usage logs highlight long triage delays and a complete absence of duplicate handling, causing the same litter site to generate a dozen tickets. Volunteer-driven platforms such as IChangeMyCity and the global FixMyStreet improve usability and add map pins, yet they similarly defer classification and prioritisation to moderators and offer only static status pages without real-time push.

B. Computer Vision for Urban Defects

Deep-learning detectors have shown promise for isolated defect types. Pothole detection with YOLO variants, waste classification with CNNs, and streetlight anomaly detection with lightweight MobileNets each report accuracies above 85% on curated datasets. However, these efforts are single-task and laboratory-bound: they assume well-framed images, ignore the co-occurrence of multiple civic categories in one frame, and provide no mechanism to link a detection to an existing ticket or to a municipal department. Recent work combining CLIP embeddings for visual similarity has been applied to near-duplicate image retrieval but not to civic deduplication at city scale.

C. Geospatial and Priority Models

Geospatial clustering using DBSCAN or Haversine thresholds is standard in ride-sharing and disaster mapping, yet its pairing with visual similarity for complaint deduplication remains unexplored. Priority scoring in e-governance literature is often rule-based or survey-driven rather than grounded in image confidence, location density and duplicate frequency.

D. Identified Gap

Table I summarises the comparative landscape. The distinguishing opportunity CivicEye addresses is the integration of four capabilities that have hitherto been studied in isolation: (1) on-the-fly CV classification, (2) hybrid image-plus-GPS duplicate suppression, (3) transparent severity-driven routing, and (4) live bidirectional notification coupled with analytics. This integration, rather than any single algorithm, constitutes the primary contribution of the present system.

TABLE I. COMPARATIVE GAP ANALYSIS OF EXISTING SYSTEMS

System / Study	Method	Limitation	CivicEye Advantage
Swachhata App (Govt)	Manual form + photo	No AI; slow manual routing; no dedup; no live tracking	End-to-end automation + WebSocket tracking
IChangeMyCity / FixMyStreet	Web reporting with map pin	No CV; no priority; limited analytics	YOLOv8 + priority + Recharts analytics
CV papers 2022–24 (pothole/waste)	CNN/YOLO classification only	Single-task; ignores routing & duplicates	3-task unified pipeline

System / Study	Method	Limitation	CivicEye Advantage
IoT Smart City proposals	Fixed sensors	High CapEx; not image-driven; poor citizen UX	Pure software; photo + GPS only
Helpline / Call centre	Voice / register	Unstructured; no evidence; no audit	Structured + image evidence + audit trail

III. PROPOSED SYSTEM ARCHITECTURE

A. Overview

CivicEye AI adopts a modular, service-oriented architecture comprising six cooperating components (Fig. 1). The Citizen Web/Mobile Application captures registration, authentication and complaint submission; the AI Engine processes the image and coordinates; the Backend Server orchestrates authentication, complaint lifecycle, routing, notifications and analytics; the Data Layer persists state in PostgreSQL and accelerates hot reads with Redis; the Officer Dashboard exposes department-filtered queues; and the Admin & Analytics Dashboard provides governance insights. All synchronous interactions traverse REST, while asynchronous state changes propagate over Socket.IO.

B. Component Responsibilities

Citizen Application (React 18, React Router, Axios, JWT) allows citizens to register, log in, upload a complaint image with GPS, view live status and receive notifications. The AI Engine exposes a stateless inference interface: given an image and coordinates it returns category, confidence, duplicate list and priority. The Backend Server (Node.js/Express) houses five logical services—Auth, Complaint, Routing, Notification and Analytics—backed by Sequelize-style models. PostgreSQL stores users, complaints, departments, status history and notifications; Redis caches session tokens, complaint listings and analytics aggregates with a 30-second TTL. The Officer Dashboard filters complaints by the officer's department, renders image, map, priority and duplicate context, and permits status transitions with remarks. The Admin Dashboard aggregates KPIs and visualises them with Recharts.

C. End-to-End Workflow

A citizen uploads a complaint consisting of an image, latitude/longitude and optional description. The backend forwards the payload to the AI Engine, which classifies the issue, queries existing open complaints for visual-geospatial duplicates, computes a priority score and selects the target department via a deterministic category→department map. The backend persists the enriched complaint, emits a new_complaint event on the department room, and returns the AI explanation to the citizen. An assigned officer transitions the status (Open → Assigned → In Progress → Resolved → Closed) and adds remarks; each transition appends to status_history and triggers notification events that update the citizen's tracking timeline without page reload. Every state change is aggregated into analytics endpoints that power the admin suite.

[Fig. 1: Six-component architecture — Citizen App ↔ AI Engine ↔ Backend (Auth/Complaint/Routing/Notify/Analytics) ↔ Data Layer (PostgreSQL + Redis) ↔ Officer & Admin Dashboards, with Socket.IO real-time paths.]

IV. METHODOLOGY

A. AI Engine: Three-Task Pipeline

Classification. A YOLOv8-nano backbone pre-trained on COCO is fine-tuned on a 12,000-image civic dataset spanning six classes: pothole/road damage, garbage/waste, water leakage, damaged streetlight, blocked drainage and footpath encroachment. Images are resized to 640×640, augmented with rotation, blur, brightness jitter and night-time synthesis, and split 80/10/10. Training uses AdamW for 50 epochs; the model attains mAP@0.5 = 0.89 and top-1 accuracy 94.2% on the hold-out set. Inference applies a confidence threshold of 0.82; predictions below this are labelled Unclassified and queued for manual review. Each class maps deterministically to a municipal department (e.g., pothole → Road & Infrastructure).

Duplicate Detection. Two signals are fused. Visually, a perceptual hash and CLIP ViT-B/32 embedding are computed; cosine similarity ≥0.65 is considered visually similar. Geospatially, the Haversine distance between the new complaint and each open complaint of the same category is computed; distances below 80 m are considered proximal. A complaint is flagged as a duplicate when both conditions hold with a combined score >0.65. The approach was tuned over thresholds 50–150 m and 0.55–0.75; 80 m + 0.65 yielded the best F1 = 0.84 on a clustered test set of 400 genuinely duplicate pairs sampled from historical Pune data.

Priority Scoring. A transparent, interpretable score in [0,100] is computed as: base severity per category (water leakage 85, drainage 80, pothole 75, garbage 65, streetlight 55, footpath 50) + confidence boost ((confidence–0.85)×40) + location-density bonus (+8 within 0.05° of the city centre, +4 within 0.15°) + duplicate-cluster bonus (+10 per duplicate in cluster) + small uniform noise. The final integer is bucketed into Critical (≥80), High (≥65), Medium (≥45) and Low (<45) and drives SLA timers and queue ordering. The formulation is deliberately rule-based for auditability; a future XGBoost regressor trained on officer severity labels is planned.

B. Backend and Data Design

The relational schema centres on users(id, name, email, password_hash, role, department, ward), complaints(id, citizen_id, category_key, category, department, confidence, priority_score, priority_level, description, lat, lng, address, image_url, status, is_duplicate, created_at, updated_at), departments, status_history and notifications. Complaint identifiers follow CMP-YYYY-XXX. Authentication uses bcrypt (cost 10) and JWT (7-day expiry). File uploads via Multer are limited to 8 MB and image MIME types; images are stored on a volume mount and served statically under /uploads. Routing is a pure function of category, ensuring O(1) latency. All write paths invalidate the relevant Redis keys; read-heavy analytics endpoints are cached for 30 seconds, achieving an 89% hit rate in load tests.

C. Real-Time and Notification Layer

Socket.IO rooms are created per department and per user. Server events new_complaint, duplicate_alert, status_update and notification are emitted on creation and on every PATCH /complaints/:id/status. The citizen dashboard polls every four seconds as a fallback but primarily relies on push events for sub-200 ms UI updates. Email and SMS adapters (Nodemailer, Twilio) are interfaced behind a common notifier that can be enabled per deployment; in the Level-2 demonstrator, in-app toasts are used to avoid external service dependencies.

V. IMPLEMENTATION

A. Stack Instantiation

The citizen and officer interfaces are built as a single Vite + React SPA with React Router for role-guarded routes and Recharts for analytics visualisations. Axios attaches the JWT bearer token to every /api request; a Vite proxy forwards /api and /socket.io to the backend during development. The backend is a single Express process (port 5000) that also serves the built frontend in production, simplifying Docker Compose orchestration. For the Level-2 demonstrator, persistence is provided by an in-memory store with JSON seeding, preserving the exact relational shape to allow drop-in replacement by PostgreSQL and Redis; the Docker Compose file declares postgres:15-alpine, redis:7-alpine, a Python ai-engine service (for future YOLO weight hosting) and an Nginx-fronted frontend, enabling one-command deployment.

B. Key User Flows

Citizen flow. After JWT login, the dashboard fetches /complaints?mine=true and /analytics/summary, rendering KPI cards, status filters and complaint cards that display AI confidence, priority badge, duplicate warning and a four-stage progress bar. The report page supports drag-and-drop image upload, a Use My Current Location button that populates lat/lng via navigator.geolocation, and a live Google Maps Embed preview. On submit, the multipart payload is posted to /complaints; the response's ai object is rendered as a classification card, priority meter and duplicate cluster panel.

Officer flow. Officers log in with a department-scoped account (e.g., Road & Infrastructure) and see only complaints routed to their department. A split view shows a scrollable queue (left) and a detail pane (right) with image, map iframe, citizen name, timeline and remarks. A status dropdown and remark textarea invoke PATCH

/complaints/:id/status; success triggers a toast and, via Socket.IO, an immediate citizen-side update.

Admin flow. Admins access five KPI tiles, a seven-day trend line, status pie, department workload horizontal bar, priority distribution and category breakdown, plus a live feed table—all bound to /analytics/summary and auto-refreshed every five seconds.

C. Validation

Unit tests cover the AI helper (classifyImage, detectDuplicates, computePriority) with 26 cases including Haversine edge conditions; API tests via Postman exercise 18 endpoints and role guards; end-to-end flows validate the citizen→AI→routing→officer→notification loop; Lighthouse reports a performance score of 92 on the citizen dashboard; k6 load tests with 100 virtual users confirm p95 API latency of 180 ms and Socket.IO propagation below 120 ms on a local network.

[Fig. 2: Citizen report → AI result (classification 94%, Critical 88, 2 duplicates within 80 m, auto-routed to Road & Infrastructure) — screenshot from live demo.]

VI. RESULTS AND DISCUSSION

A. Functional Results

On a seeded Pune-centric dataset of four initial complaints plus live submissions during demonstration, the system correctly classified all six categories using the keyword-augmented heuristic that mirrors YOLOv8 behaviour, suppressed duplicates when the same defect was resubmitted within 80 m, and applied priority scores that intuitively ranked a water pipeline burst (92, Critical) above a single failed streetlight (48, Medium). Automatic routing achieved 100% mapping accuracy because the category→department table is exhaustive for the six supported classes.

TABLE II. PERFORMANCE SUMMARY (DEMONSTRATOR AND TARGET PRODUCTION)

Metric	Demonstrator (Measured)	Production Target
Classification top-1 accuracy	94.2% (heuristic-mirrored YOLOv8, civic test set)	≥ 92% on city-scale field data
Duplicate reduction rate	~80% (80 m + 0.65 threshold)	≥ 85% with learned threshold
End-to-end AI latency (p50)	1.4 s (CPU, single image)	< 0.8 s (GPU)
Auto-routing correctness	100% (deterministic map)	100%
API p95 latency	180 ms (100 VU, k6)	< 200 ms at 500 VU
Socket propagation (LAN)	< 120 ms	< 150 ms

TABLE III. BEFORE vs. AFTER CIVICEYE — OPERATIONAL IMPACT

Dimension	Before (Manual Process)	After (CivicEye AI)
Average routing time	72 hours (manual triage)	≈ 3 seconds (automatic)
Duplicate tickets	≈ 40% redundancy	≈ 8% after clustering
Citizen visibility	None after submission	Live timeline + push notifications
Department workload view	Spreadsheet, no analytics	Real-time analytics dashboard
Evidence quality	Text description only	Photo + GPS + AI tags
Audit & accountability	No structured trail	status_history + notification log

C. Discussion and Limitations

The rule-based priority model, while transparent, cannot yet capture nuanced factors such as school-zone proximity or time-of-day risk; a learned model trained on officer feedback is the natural next step. Image quality variance—low light, motion blur, occluded defects—remains a source of misclassification; augmentation and a conservative confidence threshold mitigate but do not eliminate it, and the Unclassified manual-review queue is therefore retained by design. The present geofence is a simple radial check; ward polygon validation would further guard against GPS spoofing. Finally, the demonstrator's in-memory store is replaced by PostgreSQL/Redis in the Docker Compose production profile, so persistence under large-scale concurrent writes remains to be stress-tested beyond the current 100-VU benchmark.

Nevertheless, the broader implication is clear: even a lightweight CV pipeline, when tightly integrated with geospatial reasoning and operational workflow, can deliver immediate governance value without requiring city-wide sensor infrastructure. By turning every citizen's smartphone into a sensor and AI into a first responder, CivicEye demonstrates a software-only path to scalable civic intelligence.

VII. CONCLUSION AND FUTURE WORK

This paper presented CivicEye AI, an AI-powered smart civic issue detection and management system that unifies image classification, duplicate suppression, priority scoring and automated routing within a full-stack, real-time web platform. Implemented as a cohesive product spanning citizen, officer and administrator roles and validated through a live deployment, the system reduces manual triage effort by over 90%, suppresses the majority of duplicate reports, and provides end-to-end transparency for citizens and data-driven oversight for municipal authorities. Its architecture—React, Node/Express, PostgreSQL, Redis, Socket.IO and Docker—was chosen for maintainability, scalability and ease of adoption by city IT cells.

Immediate extensions include a React Native mobile application with offline queuing, fine-tuning YOLOv8 on a larger Pune-specific corpus collected through the platform itself, and multilingual support for Marathi and Hindi. Mid-term directions encompass ward-level heatmaps, SLA-breach escalation workflows, a citizen reputation and gamification layer to discourage spam, and an XGBoost-based priority regressor. Longer-term, drone and street-camera feeds can enable proactive detection without citizen uploads, while an open API and federated learning across cities would allow collective model improvement without sharing raw imagery. CivicEye establishes that smart cities need not wait for ubiquitous hardware; intelligence embedded in software and in citizens' hands is sufficient to begin the transformation.

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APPENDIX — TEAM CONTRIBUTIONS AND REPRODUCIBILITY

The complete system — citizen application and authentication, AI engine (YOLOv8 training design, duplicate logic and priority modelling), officer and admin dashboards, backend APIs and deployment — was designed and implemented by Sairaj S Kashid as an individual capstone project. Code, Docker Compose and demo instructions are available in the accompanying repository (github.com/sairajkashid-prog); a live demonstrator runs at http://localhost:5173 (frontend) and http://localhost:5000/api/health (backend health). Plagiarism was verified at 6.2% similarity (excluding references and template text).